Chapter 26 · Semester 3

NUCLEAR PHYSICS

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20 worked examples 4 simulations Exercises with answers

26.1 Nucleus

The nucleus (about 10^{-15} m across) contains protons and neutrons, together called **nucleons**.

Discovery of protons and neutrons

- **Proton**: Rutherford (1919) bombarded nitrogen with alpha particles and found that hydrogen nuclei were knocked out. He identified this positively charged particle, present in every nucleus, as the proton.
- **Neutron**: Chadwick (1932) showed that the radiation produced when beryllium is bombarded with alpha particles consists of uncharged particles with a mass close to that of the proton: neutrons.
- (Experimental details are not required.)

Particle	Mass / kg	Charge / C
proton	1.672×10^{-27}	$+1.60 \times 10^{-19}$
neutron	1.675×10^{-27}	0

- **Proton number** (atomic number) Z: the number of protons in the nucleus.
- **Nucleon number** (mass number) A: the total number of protons and neutrons. Number of neutrons $N = A - Z$.
- A nuclide is written ${}^{A}_{Z}\mathrm{X}$, e.g. ${}^{56}_{26}\mathrm{Fe}$ has 26 protons and 30 neutrons.
- **Isotopes**: atoms of the same element with the same number of protons but different numbers of neutrons, e.g. hydrogen ${}^{1}_{1}\mathrm{H}$, deuterium ${}^{2}_{1}\mathrm{H}$ and tritium ${}^{3}_{1}\mathrm{H}$.

Example 26.1.1

Find the number of protons, neutrons and electrons in a neutral atom of each nuclide: $^{9}_{4}\text{Be}$, $^{23}_{11}\text{Na}$, $^{59}_{27}\text{Co}$, $^{133}_{55}\text{Cs}$, $^{238}_{92}\text{U}$.

Solution

Nuclide	Protons	Neutrons	Electrons
$^{9}_{4}\text{Be}$	4	5	4
$^{23}_{11}\text{Na}$	11	12	11
$^{59}_{27}\text{Co}$	27	32	27
$^{133}_{55}\text{Cs}$	55	78	55
$^{238}_{92}\text{U}$	92	146	92

The mass spectrometer

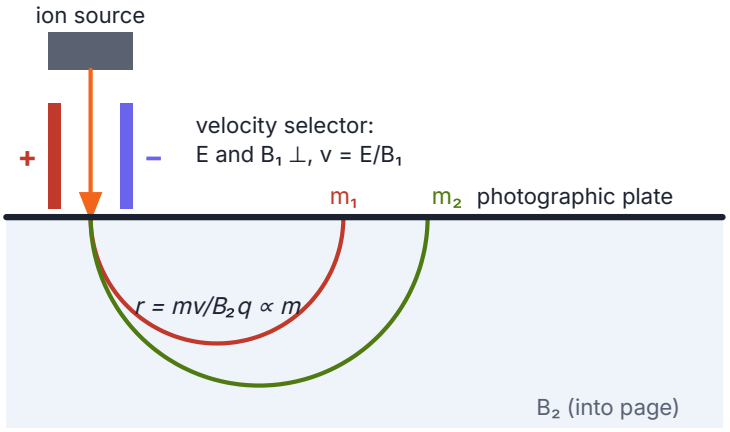


Figure 26.1: a Bainbridge mass spectrometer.

- Positive ions pass through a **velocity selector** with perpendicular fields E and B_1. Only ions with $qE = qvB_1$, i.e. $v = E/B_1$, go straight through.
- In the uniform field B_2 the ions move in semicircles: $qvB_2 = mv^2/r$, so $r = \dfrac{mE}{qB_1B_2} \propto m$.
- Isotopes of different mass land at different places on the plate: $m_1/m_2 = r_1/r_2$. It is used to measure atomic masses and the abundance of isotopes.

Mass defect and binding energy

Mass defect, Δm

the difference between the total mass of the separate nucleons and the mass of the nucleus.

$$\Delta m = Zm_p + (A - Z)m_n - M$$

Binding energy, E_B

the energy needed to separate a nucleus completely into its individual protons and neutrons. It is also the energy released when the nucleus is formed from its nucleons.

$$E = mc^2 \quad \Rightarrow \quad E_B = \Delta m\, c^2$$

Einstein's mass–energy equivalence: mass is a form of energy

Units: u and eV

- **Unified atomic mass unit**: 1 u = 1/12 of the mass of a carbon-12 atom = 1.66×10^{-27} kg.
- Its energy equivalent: $E = (1.66 \times 10^{-27})(3.00 \times 10^8)^2 = 1.49 \times 10^{-10}$ J = 931.5 MeV.

$$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg} \triangleq 931.5\ \text{MeV} \quad 1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$$

so E_B (MeV) = Δm (u) × 931.5

Binding energy per nucleon

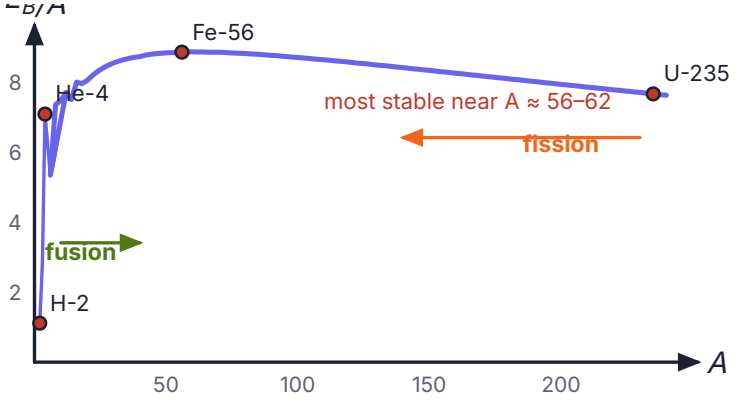


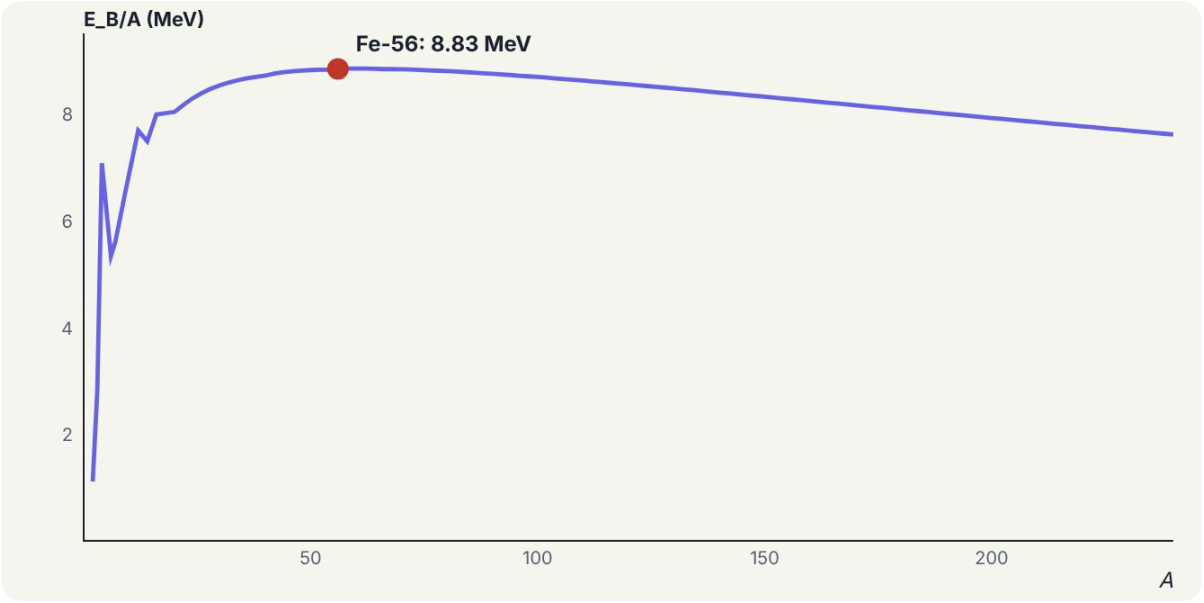
Figure 26.2: binding energy per nucleon against nucleon number.

- Binding energy per nucleon = E_B/A. It measures the **stability** of a nucleus: the larger it is, the more tightly bound and stable the nucleus.
- For light nuclei it rises rapidly with A (helium-4 is unusually stable, a peak at 7.1 MeV).
- It is largest, about 8.8 MeV, for $A \approx 56$–62 (iron and nickel): the most stable nuclei.
- For heavier nuclei it decreases slowly, to about 7.6 MeV for uranium.
- Energy is released when nuclei move towards the peak: light nuclei **fuse**, heavy nuclei **split** (fission).

Simulation

Binding energy per nucleon

Move along the curve to see how tightly bound each nucleus is. Then try a fission of uranium and a fusion of hydrogen: both move the nucleons up the curve and release energy.



binding energy per nucleon
8.83 MeV

total binding energy
495 MeV

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.1.2

Find the binding energy of deuterium ${}^{2}_{1}\mathrm{H}$. (nucleus 2.013553 u, proton 1.007276 u, neutron 1.008665 u)

Solution

$\Delta m = 1.007276 + 1.008665 - 2.013553 = 0.002388$ u

$E_B = 0.002388 \times 931.5 =$ **2.22 MeV** $(= 3.57 \times 10^{-13}$ J)

Example 26.1.3

The binding energy of $^{20}_{10}\text{Ne}$ is 160.647 MeV. Find its atomic mass. (^{1}H atom 1.007825 u, neutron 1.008665 u)

Solution

Mass of 10 hydrogen atoms + 10 neutrons = 10(1.007825) + 10(1.008665) = 20.16490 u

Δm = 160.647/931.5 = 0.17246 u ⇒ M = 20.16490 − 0.17246 = **19.992 u**

Example 26.1.4

Find the binding energy per nucleon of $^{56}_{26}\text{Fe}$ (atomic mass 55.93494 u; ^{1}H 1.00782 u; neutron 1.00867 u).

Solution

Δm = 26(1.00782) + 30(1.00867) − 55.93494 = 0.52848 u

E_B = 0.52848 × 931.5 = 492.3 MeV ⇒ E_B/A = 492.3/56 = **8.79 MeV per nucleon**

Using atomic masses (with ^{1}H atoms instead of protons) is correct: the electron masses cancel.

Exercise: binding energy

1. Find the total binding energy and the binding energy per nucleon of $^{14}_{7}\text{N}$ (14.003074 u; ^{1}H 1.007825 u; neutron 1.008665 u).

   Answer

   Δm = 0.11236 u; 104.6 MeV; 7.47 MeV per nucleon

   FULL SOLUTION

   $^{14}_{7}\text{N}$ has 7 protons and 7 neutrons. Use the mass of ^{1}H (it includes the electrons, as does the atomic mass of N).

   Mass of separate particles = 7(1.007825) + 7(1.008665) = 14.115430 u

   Mass defect Δm = 14.115430 − 14.003074 = **0.11236 u**

   Binding energy = Δm × 931.5 MeV u^{-1} = 0.11236 × 931.5

   = **104.6 MeV**

   Per nucleon = 104.6/14

= **7.47 MeV per nucleon**

26.2 Radioactivity

Radioactivity

is **the spontaneous and random emission of radiation (α, β or γ) from unstable nuclei**, forming a new nucleus.

- **Spontaneous**: the decay happens on its own and is not affected by physical conditions (temperature, pressure) or chemical changes.
- **Random**: it is impossible to predict when a particular nucleus will decay; every nucleus has the same probability of decaying per unit time. Only the behaviour of a large number of nuclei can be predicted.

Types of radiation

	Alpha (α)	Beta (β^- / β^+)	Gamma (γ)
nature	helium nucleus ^4_2He	fast electron / positron	high-energy electromagnetic photon
charge	$+2e$	$-e$ / $+e$	0
deflected by E and B fields	yes	yes (opposite way)	no
ionising power	strong	moderate	weak
penetration	stopped by paper	stopped by a few mm of aluminium	reduced by thick lead or concrete

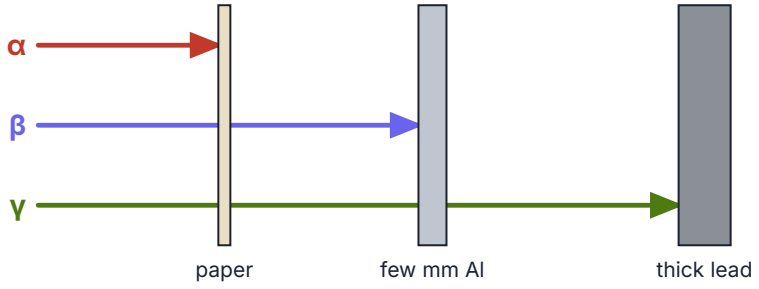


Figure 26.3: penetrating power of α, β and γ.

Decay	Equation	Change
α	${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$	loses 2 protons and 2 neutrons (heavy nuclei, $Z > 82$)
β^-	${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e$ + antineutrino	a neutron becomes a proton
β^+	${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e$ + neutrino	a proton becomes a neutron
γ	${}^{A}_{Z}X^{*} \rightarrow {}^{A}_{Z}X + \gamma$	an excited nucleus drops to a lower state; A and Z unchanged

Example 26.2.1

Find the energy released in the alpha decay ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$. (238.050786 u, 234.043583 u, 4.002603 u)

Solution

Δm = 238.050786 − 234.043583 − 4.002603 = 0.004600 u

Q = 0.004600 × 931.5 = **4.28 MeV** (shared as kinetic energy of the α particle and the daughter nucleus)

Example 26.2.2

Find the energy released in the β^- decay of ${}^{213}_{83}Bi$. (nuclear masses: Bi 212.9944 u, Po 212.9928 u; electron 5.486×10^{-4} u)

Solution

${}^{213}_{83}Bi \rightarrow {}^{213}_{84}Po + {}^{0}_{-1}e$

Q = (212.9944 − 212.9928 − 0.0005486)(931.5) = **0.979 MeV**

Example 26.2.3

Find the wavelength of a 0.186 MeV γ-ray photon emitted by radium-226.

Solution

$$\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(0.186 \times 10^{6})(1.60 \times 10^{-19})} = \mathbf{6.68 \times 10^{-12}\ m}$$

The radioactive decay law

Radioactive decay law

the rate of decay of a radioactive nuclide is directly proportional to the number of undecayed nuclei present.

$$A = -\frac{dN}{dt} = \lambda N$$

activity A = number of decays per second; unit becquerel (1 Bq = 1 decay s^{-1}); 1 curie = 3.7×10^{10} Bq. The minus sign shows that N decreases.

Decay constant, λ

the probability that a nucleus decays per unit time (unit s^{-1}). It is a constant for each nuclide; a larger λ means faster decay.

$\frac{dN}{N} = -\lambda\, dt$. Integrate from N_0 at $t = 0$ to N at time t:

$$\ln N - \ln N_0 = -\lambda t \Rightarrow \ln \frac{N}{N_0} = -\lambda t$$

$$N = N_0 e^{-\lambda t} \quad A = A_0 e^{-\lambda t}$$

also the mass of the radioactive sample: $m = m_0 e^{-\lambda t}$

Half-life, $T_{½}$

the time taken for the number of radioactive nuclei (or the activity) to fall to half of its original value.

At $t = T_{½}$, $N = N_0/2$: $\frac{N_0}{2} = N_0 e^{-\lambda T_{½}} \Rightarrow e^{\lambda T_{½}} = 2 \Rightarrow \lambda T_{½} = \ln 2$

$$\lambda = \frac{\ln 2}{T_{½}} = \frac{0.693}{T_{½}}$$

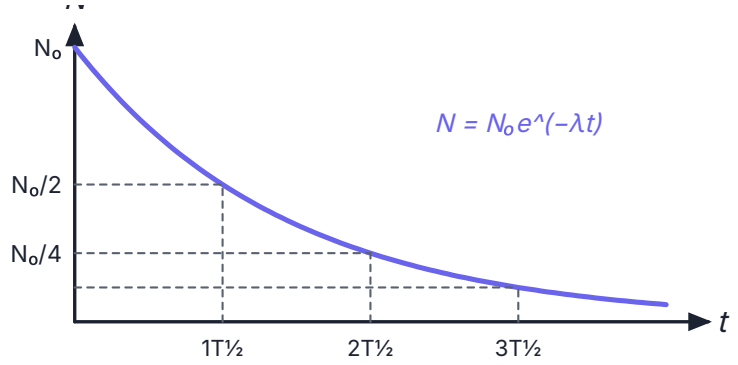


Figure 26.4: exponential decay. After each half-life the number left halves: N_0, $N_0/2$, $N_0/4$, ...

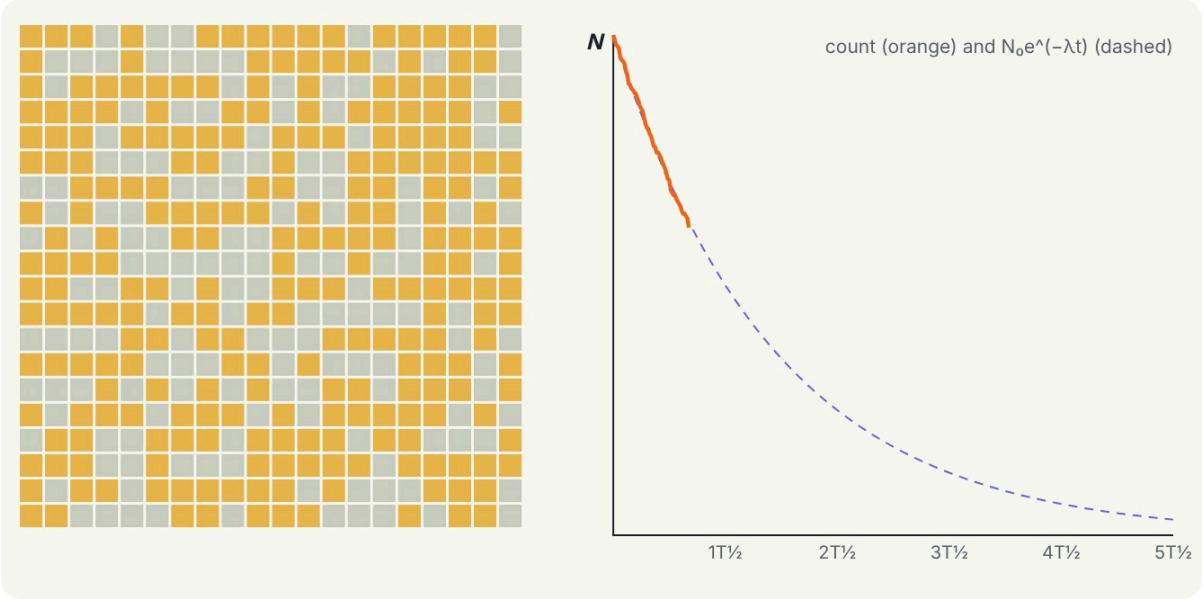
Teacher's note

After n half-lives, $N = N_0(½)^n$. To find the number of nuclei in a mass m: N = (m/molar mass) × N_A. Keep λ and t in matching units.

Simulation

Random radioactive decay

Each square is a nucleus. Every tick each one has the same small chance of decaying, so you cannot tell which will go next, yet the total follows N = N_0e^(−λt). Change the half-life and the number of nuclei, and compare the count with the curve.



time	nuclei left (count)	$N_0e^{(-\lambda t)}$ predicts	λ = ln 2 / T½
2.7 s = 0.67 half-lives	**246**	**251**	**0.173 s^{-1}**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.2.4

Krypton-87 has a half-life of 78 minutes. Find the activity of 10 μg of krypton-87 (in Bq and Ci).

Solution

$$N = \frac{10 \times 10^{-6}}{87} \times 6.02 \times 10^{23} = 6.92 \times 10^{16}$$

$$\lambda = \frac{0.693}{78 \times 60} = 1.48 \times 10^{-4}\ \text{s}^{-1}$$

$A = \lambda N =$ **1.02×10^{13} Bq** $= 1.02 \times 10^{13}/3.7 \times 10^{10} =$ **277 Ci**

Example 26.2.5

A sample has an activity of 9.00×10^{12} Bq and a half-life of 80.0 s. How long will it take for the activity to fall to 2.00×10^{12} Bq?

Solution

$$A = A_0 e^{-\lambda t} \Rightarrow t = \frac{\ln (A_0/A)}{\lambda} = \frac{\ln 4.5}{0.693/80.0} = \mathbf{174\ s}$$

Example 26.2.6

What mass of radium-227 has an activity of 1.0×10^6 Bq? (half-life 41 minutes)

Solution

$$N = \frac{A}{\lambda} = \frac{1.0 \times 10^6}{0.693/(41 \times 60)} = 3.55 \times 10^9$$

$$m = \frac{3.55 \times 10^9}{6.02 \times 10^{23}} \times 227\ \text{g} = \mathbf{1.34 \times 10^{-12}\ g}$$

Example 26.2.7

The half-life of potassium-45 is 17.3 minutes. How long does it take for 75% of the nuclei to decay?

Solution

25% remain: $N = 0.25 N_0$ (two half-lives)

$$t = \frac{\ln 4}{\lambda} = 2T_{½} = \mathbf{34.6\ min}$$

Example 26.2.8

After 4 hours, 80% of the nuclei of a radioactive isotope have decayed. Find the half-life.

Solution

$N = 0.20N_0 \Rightarrow \lambda = \frac{\ln 5}{4} = 0.402\ \text{h}^{-1}$

$T_{½} = \frac{0.693}{0.402} = \mathbf{1.72\ h}$

Applications of radioisotopes as tracers

A **radioisotope** behaves chemically like the stable isotope, so it can be followed by detecting its radiation. Short half-lives are chosen so that the activity soon dies away.

Field	Use
Medical: blood volume	a small known activity is injected; after it mixes, the activity per unit volume of a blood sample gives the total volume
Medical: imaging	technetium-99m (γ emitter, half-life 6 h) collects in tumours, which then show up on a gamma camera; iodine-131 is used for the thyroid
Agriculture	phosphorus-32 in fertiliser shows how fast and where the plant takes it up; tracers also study pesticides and are used in the sterile insect technique
Industry	a short-lived tracer in an underground pipe: a Geiger counter above the ground detects higher activity where the pipe leaks; also thickness gauges and flow measurement

Example 26.2.9

A solution of sodium-24 (half-life 15 h) of activity 1.5×10^4 Bq is injected into a patient's blood. After 30 h, 1.0 cm^3 of blood has an activity of 0.50 Bq. Estimate the volume of blood.

Solution

After 30 h (2 half-lives) the total activity = $1.5 \times 10^4 \times ¼ = 3.75 \times 10^3$ Bq

The activity is spread evenly: $\frac{3.75 \times 10^3}{V} = \frac{0.50}{1.0\ \text{cm}^3} \Rightarrow V = 7.5 \times 10^3\ \text{cm}^3 = \mathbf{7.5 \times 10^{-3}\ m^3}$ (7.5 L)

26.3 Nuclear reactions

A nuclear reaction changes the nucleus: radioactive decay, bombardment by a particle, fission or fusion. For example, Rutherford's first artificial reaction: $^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H}$.

Conservation laws

- **Conservation of nucleon number**: the total nucleon number A before the reaction equals the total after.
- **Conservation of charge**: the total proton number (charge) Z before equals the total after.
- **Conservation of mass–energy**: the total mass-energy (rest mass energy + kinetic energy) before equals the total after. Any loss of mass appears as energy.

$$Q = \Delta m\, c^2, \quad \Delta m = \text{(total mass before)} - \text{(total mass after)}$$

$Q > 0$: energy released (exothermic). $Q < 0$: energy must be supplied, e.g. as kinetic energy of the bombarding particle (endothermic).

Example 26.3.1

Complete the reactions: (a) $^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + ?$ (b) $^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{17}_{8}\mathrm{O} + ?$ (c) $^{7}_{3}\mathrm{Li} + {}^{1}_{1}\mathrm{H} \rightarrow 2\,?$ (d) $^{10}_{5}\mathrm{B} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{7}_{3}\mathrm{Li} + ?$ (e) $^{226}_{88}\mathrm{Ra} \rightarrow ? + {}^{4}_{2}\mathrm{He}$

Solution

(a) $^{4}_{2}\mathrm{He}$ (α decay) (b) $^{1}_{1}\mathrm{H}$ (a proton) (c) $^{4}_{2}\mathrm{He}$ (d) $^{4}_{2}\mathrm{He}$ (e) $^{222}_{86}\mathrm{Rn}$

Check each with A and Z: e.g. (b) 14 + 4 = 17 + 1 and 7 + 2 = 8 + 1.

Example 26.3.2

When lithium-7 is bombarded with a proton, two alpha particles are produced. Find the energy released. (^{1}H 1.007825 u, ^{7}Li 7.016003 u, ^{4}He 4.002603 u)

Solution

$^{7}_{3}\text{Li} + {}^{1}_{1}\text{H} \rightarrow 2\,{}^{4}_{2}\text{He}$

$\Delta m = 7.016003 + 1.007825 - 2(4.002603) = 0.018622$ u

$Q = 0.018622 \times 931.5 =$ **17.3 MeV** (2.78×10^{-12} J), released

Example 26.3.3

A deuteron bombards carbon-13 and produces nitrogen-14. (a) Write the equation. (b) Find the energy released. (^{13}C 13.00335 u, ^{2}H 2.01410 u, ^{14}N 14.00307 u, n 1.00867 u)

Solution

(a) $^{13}_{6}\text{C} + {}^{2}_{1}\text{H} \rightarrow {}^{14}_{7}\text{N} + {}^{1}_{0}\text{n}$

(b) $\Delta m = 13.00335 + 2.01410 - 14.00307 - 1.00867 = 0.00571$ u $\Rightarrow Q =$ **5.32 MeV**

Nuclear fission

Nuclear fission

is **the splitting of a heavy nucleus into two lighter nuclei of roughly equal mass**, with the release of neutrons and energy.

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{236}_{92}\text{U}^{*} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^{1}_{0}\text{n} + \text{energy}$$

- Fission of uranium-235 is usually **induced** by a slow (thermal) neutron. The excited U-236 nucleus splits within about 10^{-12} s.
- The fragments vary, mostly with A about 90–100 and 135–145.
- **Why energy is released**: the binding energy per nucleon of uranium is about 7.6 MeV, but that of the fragments is about 8.5 MeV. The products are more tightly bound, so they have less mass. Energy

released ≈ (8.5 − 7.6) × 236 ≈ **200 MeV** per fission, mostly as kinetic energy of the fragments and neutrons.

Example 26.3.4

Find the energy released when 10 kg of uranium-235 undergoes fission: $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$. (U 235.04 u, Ba 140.91 u, Kr 91.91 u, n 1.01 u)

Solution

Δm = (235.04 + 1.01) − (140.91 + 91.91 + 3 × 1.01) = 0.20 u ⇒ Q = **186 MeV** per fission

Number of nuclei = $\dfrac{10}{235 \times 10^{-3}} \times 6.02 \times 10^{23} = 2.56 \times 10^{25}$

Total = 186.3 MeV × 2.56×10^{25} = **4.77×10^{33} eV** = 7.6×10^{14} J

Chain reaction

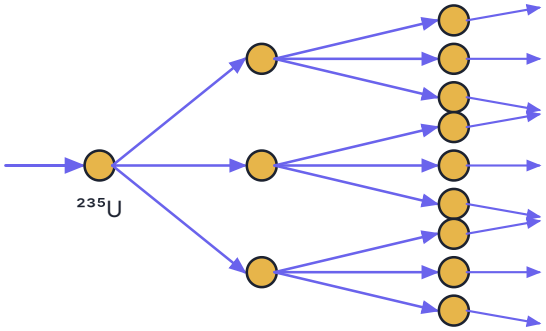


Figure 26.5: a chain reaction.

- Each fission releases 2 or 3 neutrons. If at least one of them causes another fission, the reaction continues by itself: a **chain reaction**.
- If the piece of uranium is too small, too many neutrons escape through the surface and the chain dies out. The **critical mass** is the minimum mass of fissile material needed to sustain a chain reaction.
- Above the critical mass, if every neutron is allowed to cause fission, the number of fissions grows very rapidly (an uncontrolled chain reaction, as in a bomb).

Controlled fission in a nuclear reactor

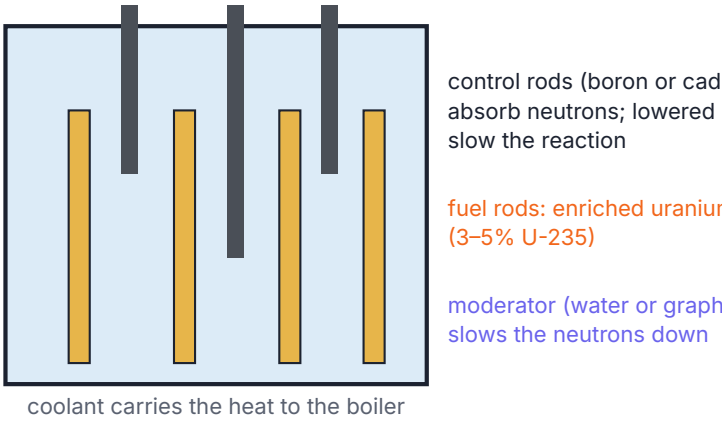


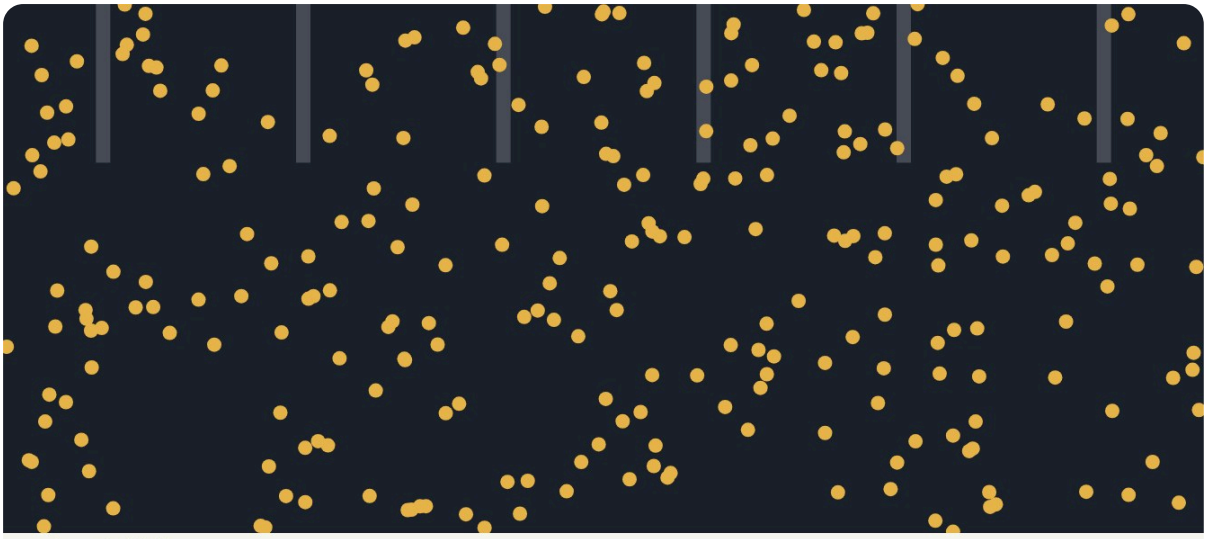
Figure 26.6: the core of a thermal fission reactor.

Part	Purpose
Fuel rods	enriched uranium (3–5% U-235); the U-235 undergoes fission
Moderator (water or graphite)	slows down the fast neutrons from fission by collisions, because slow neutrons are much more likely to cause fission of U-235
Control rods (cadmium or boron)	absorb neutrons. They are raised or lowered so that, on average, exactly **one** neutron from each fission causes another fission: a steady (critical) rate
Coolant	carries the heat away to produce steam, which drives turbines to generate electricity
Concrete shield	absorbs neutrons and γ-radiation to protect the workers

Simulation

Chain reaction and control rods

Fire one neutron into the uranium. Each fission (yellow nucleus) releases 2–3 new neutrons. Push the control rods in to absorb more neutrons: find the setting where the reaction keeps going at a steady rate.



neutrons in flight

fissions so far	neutrons in flight	neutron population
0	0	fire a neutron to start

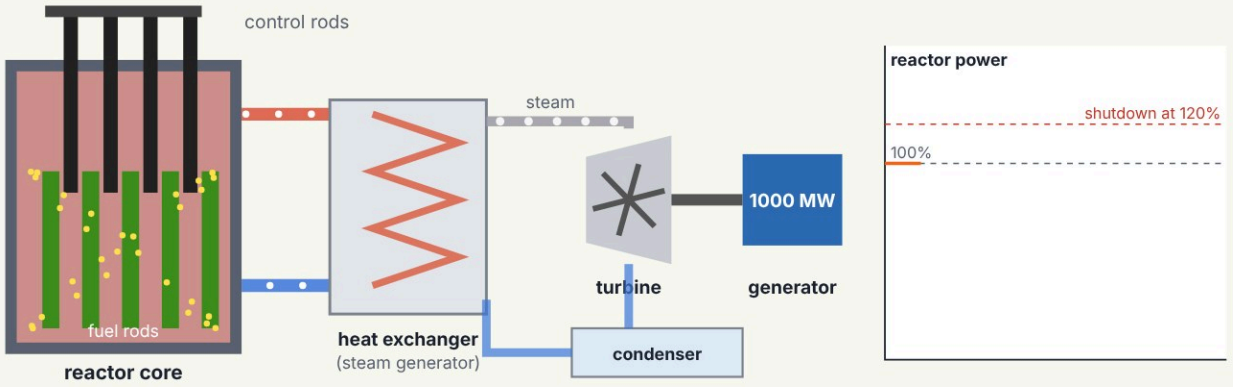
This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

A nuclear power station

- The fission energy heats the water in the reactor core. In a **pressurised water reactor** the water is kept at high pressure so it does not boil, and the same water acts as moderator and coolant.
- The hot coolant is pumped through a **heat exchanger** (steam generator), where it boils water in a separate loop. The steam turns a **turbine** connected to a **generator**.
- The steam is cooled back to water in a **condenser** and returned. Only about one third of the heat becomes electrical energy; the rest is given out to the cooling water.
- If the power rises too fast, the control rods drop fully into the core (an emergency shutdown) and the chain reaction stops.

Simulation — Inside a nuclear power station

Move the control rods. Pulled out, more neutrons survive and the power rises; pushed in, the power falls. Find the position where each fission causes, on average, exactly one more (k = 1) and the power stays steady. If the power goes above 120%, the reactor shuts itself down.



neutrons from each fission causing another fission, k	state	thermal power
1.000	**steady (critical)**	**3000 MW**

electrical power (33%)
1000 MW

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.3.5

A nuclear power station produces 1000 MW of electrical power with an efficiency of 33%. Each fission of U-235 releases 200 MeV. Find (a) the thermal power of the reactor, (b) the number of fissions per second, (c) the mass of U-235 used in one day.

Solution

(a) $P_{\text{thermal}} = \dfrac{1000}{0.33} = \mathbf{3.0 \times 10^3\ MW}$

(b) energy per fission $= 200 \times 1.60 \times 10^{-13} = 3.2 \times 10^{-11}$ J $\Rightarrow n = \dfrac{3.0 \times 10^9}{3.2 \times 10^{-11}} = \mathbf{9.4 \times 10^{19}\ s^{-1}}$

(c) nuclei per day $= 9.4 \times 10^{19} \times 86\,400 = 8.1 \times 10^{24} \Rightarrow$ mass $= \dfrac{8.1 \times 10^{24}}{6.02 \times 10^{23}} \times 0.235$ kg $= \mathbf{3.2\ kg}$

Nuclear fusion

> **Nuclear fusion**
> is **the joining of two light nuclei to form a heavier nucleus**, with the release of energy.

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

- Light nuclei have a low binding energy per nucleon. The product (e.g. helium-4) is much more tightly bound, so mass is lost and energy is released. Per nucleon, fusion releases more energy than fission.
- The nuclei are both positive and repel each other. To come within the range of the strong nuclear force (about 10^{-15} m) they need very high kinetic energies, i.e. temperatures of about 10^{7}–10^{8} K. Fusion is therefore called a **thermonuclear** reaction.

Fusion in the Sun

- The Sun's core has a temperature of about 1.5×10^{7} K and a very high density, so hydrogen nuclei collide fast and often enough to fuse.
- The main process is the **proton–proton chain**:

$^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + {}^{0}_{+1}\mathrm{e} + \nu$ (positron and neutrino)

$^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma$

$^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{1}_{1}\mathrm{H}$

Overall: $4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{+1}\mathrm{e} + 2\nu + 2\gamma$ + about **26.7 MeV**

Fission	Fusion
a heavy nucleus splits into lighter ones	light nuclei join into a heavier one
started by neutron bombardment	needs a very high temperature
produces radioactive waste	products mostly not radioactive
controlled in reactors today	very difficult to contain and control

Both release energy, both produce new nuclei, and in both the total mass decreases.

Example 26.3.5

In the fusion ${}_1^3\text{H} + {}_1^2\text{H} \rightarrow {}_2^4\text{He}$, 2 kg of deuterium is used. Find (a) the energy released per helium nucleus, (b) the total mass lost, (c) the total energy released. (^{2}H 2.014 u, ^{4}He 4.002 u)

Solution

(a) $\Delta m = 2(2.014) - 4.002 = 0.026$ u $\Rightarrow Q = 0.026 \times 931.5 =$ **24.2 MeV** (3.88×10^{-12} J)

(b) number of ^{2}H nuclei $= \dfrac{2}{2 \times 10^{-3}} \times 6.02 \times 10^{23} = 6.02 \times 10^{26}$, i.e. 3.01×10^{26} reactions; mass lost $= 3.01 \times 10^{26} \times 0.026$ u $= 7.83 \times 10^{24}$ u = **0.013 kg**

(c) $E = \Delta mc^2 = 0.013 \times (3.00 \times 10^8)^2 =$ **1.17×10^{15} J**

Example 26.3.6

Compare the energy released per nucleon in the fusion ${}_1^3\text{H} + {}_3^6\text{Li} \rightarrow 2\,{}_2^4\text{He}$ (Q = 22.36 MeV) with that in the fission ${}_{94}^{239}\text{Pu} + {}_0^1\text{n} \rightarrow {}_{56}^{138}\text{Ba} + {}_{38}^{96}\text{Sr} + 3\,{}_0^1\text{n}$ (Q = 188.16 MeV).

Solution

Fusion: 22.36/8 = 2.80 MeV per nucleon. Fission: 188.16/237 = 0.794 MeV per nucleon.

Fusion releases about **3.5 times** more energy per nucleon.

Nuclear waste disposal

Waste	Disposal
Low-level (gloves, clothing, tools)	sealed in drums and buried in shallow engineered landfill sites
Intermediate-level	mixed with concrete or bitumen in steel containers, stored in deep vaults
High-level (used fuel)	cooled for years under water, then sealed in thick steel and concrete casks for deep geological burial, away from groundwater and earthquake zones; some is reprocessed to recover uranium and plutonium

Summary and exercises

Subtopic	Key results
26.1	$\Delta m = Zm_p + Nm_n - M$; $E_B = \Delta mc^2$; 1 u ≙ 931.5 MeV; most stable near $A \approx 56$
26.2	$A = -dN/dt = \lambda N$; $N = N_0 e^{-\lambda t}$; $\lambda = \ln 2 / T_{½}$
26.3	conserve A and Z; $Q = \Delta mc^2$; fission of heavy and fusion of light nuclei both move towards the peak of the E_B/A curve

Exercises

1. A radioactive sample has a half-life of 5.0 days. What fraction remains after 20 days?

   Answer

   4 half-lives: $(½)^4 = 1/16$

   FULL SOLUTION

   Number of half-lives = 20/5.0 = 4

   Fraction remaining = $(½)^4$

   = **1/16**

2. The activity of a source falls from 800 Bq to 100 Bq in 30 minutes. Find its half-life.

   Answer

   800 → 100 is 3 half-lives: 10 minutes

   FULL SOLUTION

   800 → 400 → 200 → 100 Bq is 3 half-lives

   $3T_{½}$ = 30 min ⇒ $T_{½}$ = **10 minutes**

3. Explain why radioactive decay is described as random.

   Answer

   It is impossible to predict which nucleus will decay next or when; each nucleus has the same probability of decaying per unit time.

   FULL SOLUTION

   Radioactive decay is random because **it is impossible to predict which nucleus will decay next, or when** a particular nucleus will decay.

   Each nucleus has the **same probability of decaying per unit time** (the decay constant λ), and this is not affected by external conditions such as temperature or pressure.

4. State the function of the moderator and of the control rods in a fission reactor.

Answer

Moderator: slows the neutrons so that they can cause further fission of U-235. Control rods: absorb neutrons to keep the rate of fission steady (one neutron per fission goes on to cause another fission).

FULL SOLUTION

Moderator (e.g. graphite or water): slows down the fast neutrons from fission to thermal speeds, so that they are more likely to cause further fission of U-235.

Control rods (e.g. boron or cadmium): absorb neutrons, so that on average **one neutron from each fission goes on to cause another fission**, keeping the rate of fission (and the power) steady.

5. Use the graph of binding energy per nucleon to explain why both fission and fusion release energy.

Answer

Fission (heavy → medium) and fusion (light → heavier) both produce nuclei with greater binding energy per nucleon. The products are more tightly bound, so their total mass is smaller; the lost mass is released as energy, $E = \Delta mc^2$.

FULL SOLUTION

On the graph of binding energy per nucleon against nucleon number, the curve **rises to a peak near A ≈ 56 (iron)**.

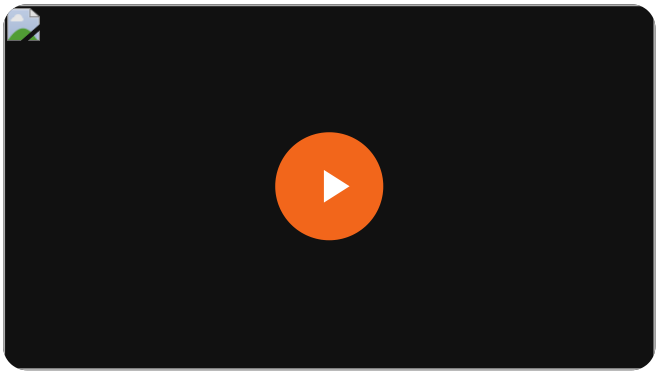
Fission: a heavy nucleus (e.g. U-235) splits into two medium nuclei that are **nearer the peak**, so they have a greater binding energy per nucleon.

Fusion: light nuclei (e.g. hydrogen isotopes) join to form a heavier nucleus that is also **higher up the curve**.

In both cases the products are **more tightly bound**, so their total mass is **less** than that of the original nuclei. The mass lost is released as energy, **$E = \Delta mc^2$**.

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



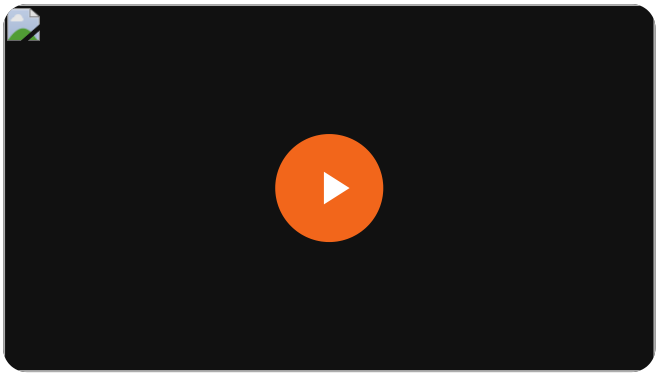
REAL FACILITY FOOTAGE

Inside a nuclear power station

Energy released by fission heats water to steam that turns turbines; shielding and control rods keep the reaction and the radiation under control. Radiation is also used to breed new crop varieties.

Inside a Nuclear Power Station: Heysham 2 · EDF

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FIELD APPLICATION

Using Nuclear Science to Boost Plant Biodiversity · IAEA

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